

# **A Comparison of Classical versus Machine Learning Models for Demand Forecasting and their Impact on Operational Outcomes**

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## **ABSTRACT**

The application of time series forecasting to predict future values of a given variable is a necessary task for any entity that operates in the sales space. Manufacturing and retail both rely on this technique to plan production schedules and adjust inventory levels in a manner that both satisfies demand and reduces operational costs. Although accurate forecasts can be deemed a necessity, the optimal methodology for producing them can be obscured by the extensive choice of modeling techniques. The following paper details a comparative case study on classical (SARIMA and Holt-Winters) versus machine learning (XGBoost and Temporal Convolutional Networks) approaches to demand forecasting. It captures key data preprocessing steps, modeling considerations, forecasting performance, and impacts on relevant business KPIs using publicly available datasets on alcohol sales and other predictor variables.

## **I. PROBLEM DEFINITION**

Whether it be a manufacturer, retailer, or distributor, every entity that deals in selling products or offering a service must be mindful of future demand to adjust labor schedules and inventory accordingly. Doing so ineffectively can lead to lost sales and customer dissatisfaction, stagnant inventory that eats up valuable warehousing space, and an overall decline in operational efficiencies and company profitability. To combat these potential scenarios, companies often adopt a systematic method for predicting future sales, typically known as demand forecasting.

Despite every demand forecasting system sharing the same goal of producing the most accurate and beneficial forecasts, their iterations throughout history, and even still today, can usually be seen to adopt various and distinct methodologies for arriving at their final sales estimates. Inputs for these systems range from purely qualitative, derived from subject matter experts, to purely quantitative, with hybrid systems existing in the middle. There exist differences even within these defined frameworks as well. Focusing on the purely quantitative systems, some entities may rely on traditional univariate statistical models such as ARIMA, while others have begun to utilize the more modern machine learning frameworks. With so many unique ways of forecasting demand, it begs the question of which method is the most successful and under what circumstances.

The following report describes a comparative case study on classical, univariate approaches to demand forecasting with machine learning models. It highlights assumptions, necessary preprocessing steps, and modeling considerations for ARIMA/SARIMA, Holt-Winters, XGBoost, and Temporal Convolutional Networks. In addition to a comparison of forecasting performance, this study includes an analysis of the operational impacts of such models using key business KPIs relevant in the manufacturing and retail industries.

## **II. DATA COLLECTION, CLEANING, AND PREPROCESSING**

This study utilized five distinct datasets for the purpose of time series forecasting. The main dataset contains sales/demand data and was sourced from an alcohol distributor operating in Montgomery County, Maryland. It possesses SKU-level sales figures by calendar month, in addition to product characteristics, and the source of the sales. For the machine learning models, four additional datasets were brought in to provide supplemental predictor variables. These include data on CPI, employment numbers, and weather, all specific to Maryland and sourced from public, federal websites. All five datasets were ingested via .csv files and transformed into Pandas dataframes for later cleaning and preprocessing. Initial

dataset characteristics are listed in Figure 1, with missing values outside of the demand date range indicated as OB\* (out-of-bounds).

| Dataset        | Shape (c, r) | Missing Values | Date Range       | Date Interval |
|----------------|--------------|----------------|------------------|---------------|
| Alcohol Demand | 9, 307,720   | 171            | 6/2017 – 9/2020  | Monthly       |
| All CPI        | 16, 11       | 3 (OB*)        | 1/2015 – 12/2025 | Monthly       |
| Alcohol CPI    | 16, 11       | 3 (OB*)        | 1/2015 – 12/2025 | Monthly       |
| Employment     | 8, 131       | 6 (OB*)        | 1/2015 – 11/2025 | Monthly       |
| Weather        | 7, 108       | 0              | 1/2017 – 12/2025 | Monthly       |

Figure 1. Initial Dataset Profiles

Prior to handling missing data within the demand dataset, the fields irrelevant to the study were removed or altered. This included removing accessory product characteristic fields and aggregating the two different sales columns (retail and warehouse) to form a singular “Sales” value for the respective month. This step was performed first to avoid the unnecessary dropping of data points due to missing data unrelated to the study. Each dataset was then evaluated for missing values. The supporting variable datasets were whole in terms of having a value for each respective period. One month from the demand dataset was missing sales data in its entirety, and the handling of this will be discussed later in the Exploratory Data Analysis section.

After observing the number of different SKUs (over 30,000) and the tendency of low-volume products to have sparse sales data across the entire time horizon, the demand dataset was processed in a way to both ensure data availability and promote a digestible scope size for the study. Rows with SKUs not belonging to the main four product categories were dropped, and the remaining sales data was aggregated up to the product category level. The four categories include: wine, beer, liquor, and nonalcoholic products. This resulted in a singular sales value per month for each individual category, resulting in four distinct time series for forecasting evaluation.

As the classical modeling approaches used a univariate approach, copies of the four individual time series were set aside for later EDA. For the multivariate machine learning models, each supporting dataset was aligned appropriately on a monthly index with each predictor variable being represented as an individual column. The supporting datasets were then joined to each of the four product category time series, resulting in each row having a respective sales value, as well as a value for each supporting predictor variable. The predictor and target variable were all quantitative in nature and were coerced to be of type float.

### III. EXPLORATORY DATA ANALYSIS AND FEATURE ENGINEERING

The demand dataset consisted of monthly figures ranging from 06/2017 to 09/2020, constituting 40 periods of sales figures per product category. The supporting datasets contained data for each of these periods, and out-of-bound months were trimmed to properly align the dataframes.

It was instrumental that the nature of the time series be characterized first, specifically looking for evidence of trend and seasonality. The presence of seasonality, trend, and the magnitude of any potential seasonality would provide the information necessary to decide on either a non-seasonal ARIMA or seasonal SARIMA model, as well as if the data was stationary (a major assumption of the ARIMA/SARIMA models). Time series decompositions and line graphs of demand over time were generated as preliminary evidence. From these, it was evident that the different product categories possessed a seasonal element, that is, sales were noticeably dependent on the time of year. There also appeared to be no overarching trend in sales, that is, sales YoY did not appear to increase or decrease. The distinct seasonal periods and lack of trends can be seen in Figure 2.

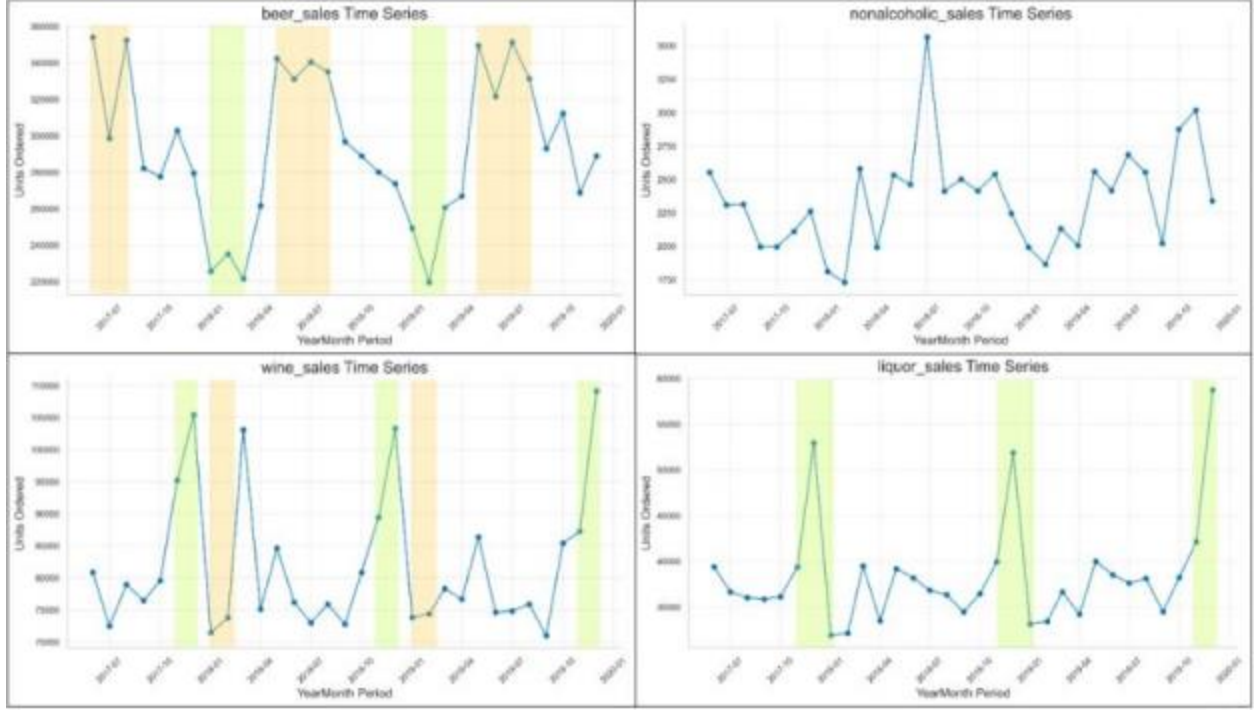


Figure 2. Demand time series with highlighted seasonal periods

This suggested that the seasonal variation of ARIMA, SARIMA, would be the best fit for time series forecasting on these datasets. The presence of a seasonal component also provided the information necessary to correctly impute the missing sales data on 08/2020. The empty month was imputed with the lag12 sales figure, which is the value of the same month, 12 months back. This was done to retain the seasonal component within each product category.

Determining if the time series were stationary required a more nuanced analysis that depended on various analytical techniques, including visual assessment, auto- and partial autocorrelation plots, as well as hypothesis tests. Although the time series did exhibit a strong seasonal component, the magnitude of seasonal periods appeared constant over time. This, along with no evidence of a clear trend, suggested that the autocorrelation plots and hypothesis tests were necessary. ACF plots also failed to suggest non-stationarity as the autocorrelations for subsequent lags did not show a slow decay typical of non-stationary data (Siddique et al., 2025). These findings were confirmed through both the Augmented Dickey-Fuller (ADF) and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) hypothesis tests, with both tests suggesting that the product category time series were all stationary.

As XGBoost models are not inherently designed to handle the ordered nature of time series, it was necessary to engineer predictor variables that aimed to preserve the temporal ordering and enrich the forecasts. Common in many forecasting applications, different magnitudes of lag features were created to capture the demand  $X$  months in the past, where  $X$  is the magnitude of the lag feature (Surakhi et al., 2021).

It was also important to retain the cyclical nature of time within the ML dataset. Rather than creating twelve new predictors to one-hot encode the respective month or eliminating the cyclical structure by using the number of the month, sine and cosine functions were implemented (A, S. et al., 2025). The only other novel predictor engineered was the number of national holidays for the respective month. This was thought to be a potentially influential feature as alcohol sales tend to increase during holiday periods.

#### IV. MODEL DEVELOPMENT AND EVALUATION

With the exploratory data analysis portion of this study confirming that the time series exhibited seasonality, the final selection of forecasting models included: SARIMA, Holt-Winters, XGBoost, and Temporal Convolutional Networks (TCN). For the model building portion of this study, a train:test split of 31:9 was used to ensure at least three quarters worth of forecast data were generated for evaluation. For the datasets with lagged features, and as a result required the truncation of rows equal to the max lag period, the training set was reduced accordingly.

The most appropriate hyperparameter space for each model and time series pairing was evaluated using a combination of grid search and TimeSeriesSplit to preserve the temporal ordering of the data. For the SARIMA models, the hyperparameter space available to grid search was able to be curated, and the overall search space reduced based on the associated ACF and PACF plots. The lists of tunable parameters are listed in Figure 3.

| Model        | Parameters Tuned with GridSearch                                         |
|--------------|--------------------------------------------------------------------------|
| SARIMA       | p, d, q, P, D, Q                                                         |
| Holt-Winters | alpha, beta, gamma                                                       |
| XGBoost      | learning rate, n_estimators, max_depth                                   |
| TCN          | number of channels, kernel size, dropout, learning rate, sequence length |

Figure 3. Models and the parameter space used in GridSearch

With the final hyperparameter spaces, the models were then retrained on the full training set prior to forecasting the test sets. To simulate a real-world demand forecasting scenario, a one-step forecasting approach was utilized. After predicting one period into the future, the model was then retrained on the expanded training set which now included the true value of the period just forecasted. Although a more computationally expensive approach, it allows the model to capture the most information by retraining on the largest possible training set.

Due to the small size of the dataset, the TCN models consistently failed to converge and generate consistent, accurate forecasts. To assist with model training, a separate dataset was generated for each product category that used back-imputation to generate an additional 90 periods of synthetic historical data. The back-imputed dataset results for TCN will be used in the discussion of results, but considerations will be noted as they rely on synthetic data.

#### V. RESULTS, INSIGHTS, AND RECOMMENDATIONS

##### Forecast Evaluation

RMSE and MAPE values were calculated for the forecasts generated by the optimal model configuration for each product category. RMSE was chosen as values share the same unit of measure as the time series, and MAPE was included to compare forecasting across product categories as it's a percentage-based error measure.

Across both metrics, the classical forecasting models consistently outperformed their machine learning counterparts, with Holt-Winters performing the best across three of the four categories. TCN did perform the best in the nonalcoholic category, but this was dependent on the back-imputed dataset that used synthetic historical data. The XGBoost models performed similar to the classical approaches in terms of RMSE and MAPE.

Looking at the nonalcoholic product category, each model demonstrated a noticeably higher MAPE when compared to the other beverage types. This demonstrates that relatively, all four modeling frameworks performed the worst on forecasting a time series with a less prominent seasonal component.

| Product Category    | Model         | Minimum RMSE | Minimum MAPE |
|---------------------|---------------|--------------|--------------|
| <b>Liquor</b>       | Holt-Winters  | 1,518.2      | 3.1          |
|                     | SARIMA        | 1,557.3      | 3.4          |
|                     | XGBoost       | 2,670.8      | 5.5          |
|                     | TCN (imputed) | 3,569.7      | 7.5          |
| <b>Wine</b>         | Holt-Winters  | 2,968.6      | 3.3          |
|                     | SARIMA        | 4,248.1      | 4.1          |
|                     | XGBoost       | 4,731.3      | 4.6          |
|                     | TCN (imputed) | 7,991.2      | 8.5          |
| <b>Beer</b>         | SARIMA        | 26,330.1     | 6.9          |
|                     | Holt-Winters  | 29,725.4     | 7.3          |
|                     | TCN (imputed) | 26,950.4     | 7.9          |
|                     | XGBoost       | 33,784.6     | 8.7          |
| <b>Nonalcoholic</b> | TCN (imputed) | 933.8        | 18.3         |
|                     | Holt-Winters  | 1,035.5      | 19.3         |
|                     | XGBoost       | 1,215.0      | 20.7         |
|                     | SARIMA        | 1,317.9      | 25.8         |

Figure 4. RMSE and MAPE of the best performing models for each category

#### Error Analysis

By plotting the forecast errors of the best-performing model in each category, it is possible to characterize the bias of the estimates. The tendency to over- or under-predict demand is significant in this context as each tendency impacts a different operational metric. As seen in Figure 6, the model for wine and beer displays a consistent pattern of under-forecasting, which would negatively impact fill rate and subsequently customer satisfaction. The model for liquor tends to over-forecast demand, which would result in excess inventory and the potential to raise operational and warehousing costs. The nonalcoholic forecasts had a mean approaching zero if the demand spike in 07/2020 is neglected.

| Model Type   | Beer   | Liquor | Nonalcoholic | Wine |
|--------------|--------|--------|--------------|------|
| Holt-Winters | -3,966 | 100    | -77          | -338 |
| SARIMA       | -3,620 | -187   | -12          | -756 |
| TCN          | -740   | -155   | -52          | -61  |
| XGBoost      | -2,251 | 143    | -68          | -361 |

Figure 5. Average error bias (in sales units) for each model and category



Figure 6. Error bias for best performing models

### Feature Importance

By utilizing the best-performing XGBoost model for each category, we were able to gain insight into which predictor variables were the most useful for forecasting by measuring the respective gain. Pictured in Figure 7 are the gain values of each predictor variable for the nonalcoholic XGBoost model. The 12-month lag, or the demand value one year in the past, was the most important feature for improving forecasting accuracy by an exceedingly large margin, even in the nonalcoholic time series which lacked a prominent seasonal dependence. This demonstrates the importance of creating lagged features and the benefits they provide in forecasting.

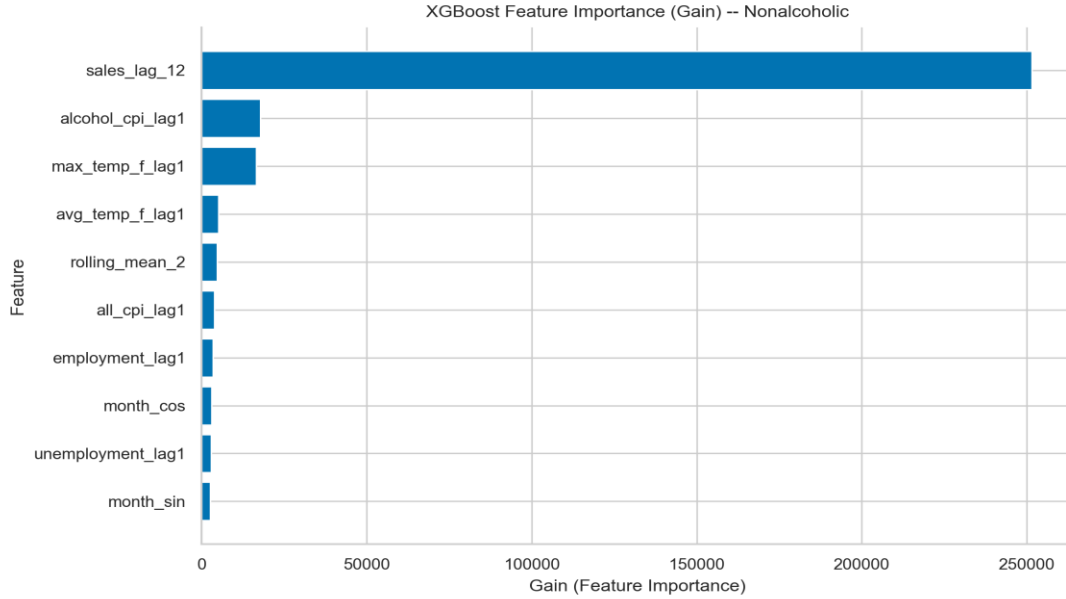


Figure 7. Feature gain derived from XGBoost

#### Operational Impact

Although metrics like MAPE and RMSE are useful for determining the objective forecasting performance on unseen data, it can fail to capture the true impact on the business. Figure 8 captures how the different forecasting models impact Fill Rate and Carrying Capacity/Inventory Levels, both key operational KPIs used within the industry. Despite TCN failing to match the MAPE and RMSE of the classical models, it performs slightly better in fill rate across the four product categories. Although the average monthly inventory levels are slightly higher, some entities might prioritize fill rate and customer satisfaction over operational costs.

| Product Cat. | Beer      |                       | Liquor    |                       | Nonalcoholic |                       | Wine      |                       |
|--------------|-----------|-----------------------|-----------|-----------------------|--------------|-----------------------|-----------|-----------------------|
| Model Type   | Fill Rate | Avg Monthly Inventory | Fill Rate | Avg Monthly Inventory | Fill Rate    | Avg Monthly Inventory | Fill Rate | Avg Monthly Inventory |
| Holt-Winters | 93%       | 296,202               | 99%       | 38,190                | 83%          | 2,632                 | 97%       | 77,740                |
| SARIMA       | 94%       | 297,740               | 97%       | 36,918                | 84%          | 2,921                 | 96%       | 75,883                |
| TCN          | 95%       | 306,990               | 93%       | 35,607                | 92%          | 3,138                 | 97%       | 77,666                |
| XGBoost      | 93%       | 295,876               | 98%       | 38,778                | 72%          | 2,141                 | 95%       | 75,618                |

Figure 8. Key operational KPIs

## VI. CONCLUSION AND FUTURE DIRECTIONS

This study aimed to compare classical approaches to time series forecasting with more modern machine learning methodologies. Forecasting performance was considered both in the context of errors and how they impacted operational KPIs downstream. Using the RMSE and MAPE from forecasts made on the original datasets (excludes the back-imputed set used for TCN convergence), the classical models of SARIMA and Holt-Winters performed the best across all four product categories. Despite XGBoost performing comparatively, these results were initially surprising as the ML models had access to an additional set of predictor variables that were thought to be relevant to the sale of alcoholic products. After more consideration, and the fact that ML forecasting accuracy drastically improved with addition of back-imputed synthetic data, the discrepancy can likely be attributed to the limited data set size. This

suggests that traditional, univariate forecasting models may serve as the best option for groups with limited historical demand data.

It is also important to call out the decoupling of error performance and impact on operational KPIs. Despite performing the worst in terms of RMSE and MAPE, the forecasts from TCN resulted in the most optimal levels of Fill Rate. This finding highlights the importance of measuring forecasting models not only with traditional statistical metrics, but also to evaluate how their implementation effects business processes and performance.

While being the most limiting factor in this study, the size of the data set and its significance in forecasting performance present itself as an interesting topic to explore further. In this study, one with a data set containing only 31 training data points, the classical univariate models excelled in terms of forecast accuracy and ease of use. Exploring forecast performance as a function of training set size could pose an interesting topic to explore as it might provide insights as to when a company should transition to machine learning models.



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